

TONGRUI (JEREMY) QIAO

Saint John, NB

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Education

University of New Brunswick – Saint John

Expected April 2027

Bachelor of Science in Computer Science

Saint John, NB

• **Dean's List** – 2025/2026 (GPA: 3.8/4.3)

Technical Skills

Programming: Python, C, C++, Java, SQL, PHP, R

AI & Data: NLP, RAG, Reinforcement Learning, PyTorch, NumPy, Pandas, Vector Search

Web & Systems: HTML, CSS, JavaScript, MongoDB, Linux, Git

Experience

Software Engineer

Apr 2026 – Aug 2026

Inteplast Group

Saint John, NB

- Built Python automation pipelines that generate technical documentation for industrial bag machine systems (RZFD series), cutting manual authoring time significantly.
- Authored LaTeX technical manuals with hyperlink navigation and QR codes linking machinery to digital operator resources.
- Wrote XML/OOXML processing scripts to convert legacy Word training checklists into structured, version-controlled formats.

Undergraduate Teaching Assistant

Jan 2025 – Present

University of New Brunswick

Saint John, NB

- Graded assignments and supported first- and second-year students in introductory programming courses.

Undergraduate Tutor

Jul 2024 – Apr 2026

The Flora Beckett Math and Science Help Centre

Saint John, NB

- Tutored mathematics, statistics, and computer science, explaining concepts and guiding students through debugging and problem-solving.

Projects

Saint John Regional Hospital ER Pre-Triage System — Python, Streamlit, LLM API, UML, Jira

Jun 2026

- Led requirements and design modelling for a web-based AI pre-triage system for two Saint John emergency rooms, producing scenario-based swimlane and UML use case diagrams for QR on-site check-in, AI triage, and live queue display.
- Designed a layered call-and-return architecture and four core data structures (patient record, triage report, queue entry, ER assignment) to keep intake, triage, and queueing loosely coupled across both sites.
- Specified a structured-JSON LLM interface returning Canadian Triage and Acuity Scale (CTAS) levels 1–5 with clinical summaries and red-flag warnings, always gated behind mandatory physician confirmation.

Ontology-Guided Occupational Disease QA System — Python, RAG, NLP, LLM

Apr 2026

- Built an ontology-guided Retrieval-Augmented Generation system to answer occupational disease questions from the ILO Diagnostic and Exposure Criteria for Occupational Diseases, enabling evidence-based responses over a large medical reference document.
- Implemented the semantic retrieval pipeline end to end — document chunking, embedding generation, and vector similarity search — to surface supporting evidence before LLM generation, improving answer relevance and reducing unsupported claims.
- Constructed a structured evaluation dataset from expert-designed questions provided by a Dalhousie University medical specialist, with baseline responses generated through Illinois Chat, a research LLM platform from the University of Illinois Urbana-Champaign.
- Designed a comparative framework across baseline, constrained-prompting, and ontology-guided RAG approaches, and performed error analysis on hallucinated content, missing evidence, and incorrect ICD-10 classifications.

Intelligent Tic-Tac-Toe Agent — Python, AI Search, CNN, DNN

Jan 2026

- Implemented minimax with alpha-beta pruning ($O(b^m) \rightarrow O(b^{m/2})$) plus depth-limited heuristics scoring winning sequences, opponent threats, and positional advantage on $n \times n$ boards.
- Trained a CNN evaluation function on self-play data (86% validation accuracy) and a DNN policy network predicting moves directly, removing tree search at the cost of optimality.

Licenses & Certifications

Generative AI Fundamentals — Databricks (Jun 2025) Microsoft AI & ML Engineering — Microsoft (Jun 2025) AI For Everyone — DeepLearning.AI (Aug 2025)